

INFT-1205 – Data Communications & Networking II

Capstone Project

VLAN Documentation

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By: Group 1

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Document Overview

This document serves as brief documentation on VLAN, trunking, EtherChannel, and inter-VLAN routing on the Datacore Solutions Devices.

1. show vlan brief – FL1

All 10 VLANs configured on FL1, and ports assigned to IT and Customer Support VLANs.

```
FL1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	IT	active	Gi1/0/1
20	Customer_Support	active	Gi1/0/3
30	Sales_and_Marketing	active	
40	HR	active	
50	Management	active	
60	System_Admins	active	
70	Accounting	active	
80	Legal	active	
999	Parking_Lot	active	Gi1/0/4, Gi1/0/8, Gi1/0/9 Gi1/0/10, Gi1/0/11, Gi1/0/12 Gi1/0/13, Gi1/0/14, Gi1/0/15 Gi1/0/16, Gi1/0/17, Gi1/0/18 Gi1/0/19, Gi1/0/20, Gi1/0/21 Gi1/0/22, Gi1/0/23, Gi1/0/24 Gi1/0/25, Gi1/0/26, Gi1/0/27 Gi1/0/28
1000	Native	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
VLAN	Name	Status	Ports
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

2. show vlan brief – FL2

All 10 VLANs configured on FL2, and ports assigned to HR and Management VLANs.

```
FL2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	IT	active	
20	Customer_Support	active	
30	Sales_and_Marketing	active	
40	HR	active	Gi1/0/1
50	Management	active	Gi1/0/3
60	System_Admins	active	
70	Accounting	active	
80	Legal	active	
999	Parking_Lot	active	Gi1/0/2, Gi1/0/4, Gi1/0/8 Gi1/0/9, Gi1/0/10, Gi1/0/11 Gi1/0/12, Gi1/0/13, Gi1/0/14 Gi1/0/15, Gi1/0/16, Gi1/0/17 Gi1/0/18, Gi1/0/19, Gi1/0/20 Gi1/0/21, Gi1/0/22, Gi1/0/23 Gi1/0/24, Gi1/1/1, Gi1/1/2 Gi1/1/3, Gi1/1/4, Ap1/0/1
1000	Native	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	

3. show interfaces trunk – FL1

Trunking to Datacore Router via Gi1/0/2 and FL2 via Po1. The native VLAN is 1000, and VLANs 10, 20, 30, 40, 50, 60, 70, 80, 1000 are allowed on the trunks.

```
FL1#show int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi1/0/2	on	802.1q	trunking	1000
Po1	on	802.1q	trunking	1000
Port	Vlans allowed on trunk			
Gi1/0/2	10,20,30,40,50,60,70,80,1000			
Po1	10,20,30,40,50,60,70,80,1000			
Port	Vlans allowed and active in management domain			
Gi1/0/2	10,20,30,40,50,60,70,80,1000			
Po1	10,20,30,40,50,60,70,80,1000			
Port	Vlans in spanning tree forwarding state and not pruned			
Gi1/0/2	10,20,30,40,50,60,70,80,1000			
Po1	10,20,30,40,50,60,70,80,1000			

4. show interfaces trunk – FL2

Trunking to FL1 via Po1. The native VLAN is 1000, and VLANs 10, 20, 30, 40, 50, 60, 70, 80, 1000 are allowed on the trunk.

```
FL2#show int trunk

Port      Mode      Encapsulation  Status        Native vlan
Po1       on        802.1q         trunking      1000

Port      Vlans allowed on trunk
Po1       10,20,30,40,50,60,70,80,1000

Port      Vlans allowed and active in management domain
Po1       10,20,30,40,50,60,70,80,1000

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,40,50,60,70,80,1000
```

5. show etherchannel summary – FL1

Port-channel 1 is in use, uses LACP with 3 ports.

```
FL1#show etherchannel summary
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Gi1/0/5(P) Gi1/0/6(P) Gi1/0/7(P)
```

6. show etherchannel summary – FL2

Port-channel 1 is in use, uses LACP with 3 ports.

```
FL2#show etherchannel summary
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Gi1/0/5(P)  Gi1/0/6(P)
                        Gi1/0/7(P)
```

7. show ip interface brief – DatacoreRouter

All the sub interfaces are up/up. Proper IP addresses are assigned as default gateways for the VLANs.

```
DatacoreRouter#show ip int br
Interface                               IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0                   unassigned      YES unset    up          up
Gi0/0/0.10                             10.160.0.1      YES manual   up          up
Gi0/0/0.20                             10.160.0.129    YES manual   up          up
Gi0/0/0.30                             10.160.0.193    YES manual   up          up
Gi0/0/0.40                             10.160.1.1      YES manual   up          up
Gi0/0/0.50                             10.160.1.193    YES manual   up          up
Gi0/0/0.60                             10.160.1.65     YES manual   up          up
Gi0/0/0.70                             10.160.1.129    YES manual   up          up
Gi0/0/0.80                             10.160.1.225    YES manual   up          up
Gi0/0/0.1000                           unassigned      YES unset    up          up
GigabitEthernet0/0/1                   unassigned      YES unset    down        down
GigabitEthernet0/0/2                   unassigned      YES unset    down        down
GigabitEthernet0/0/3                   unassigned      YES unset    down        down
Serial0/1/0                             101.1.1.1       YES manual   up          up
Serial0/1/1                             101.1.2.1       YES manual   up          up
```